ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F-ratio
Between treatments	9.9	$3 - 1 = 2$	$\frac{9.9}{2} = 4.95$	$F_0 = \frac{4.95}{6.97} = 0.710$
Error	104.6	15	$\frac{104.6}{15} = 6.97$	
Total		$n - 1 = 18 - 1 = 17$		

Step 6 : Critical value

The critical value = $f_{(15, 2),0.05} = 3.6823$.

Step 7 : Decision

As $F_0 = 0.710 < f_{(15, 2),0.05} = 3.6823$, null hypothesis is not rejected. There is no enough evidence to reject the null hypothesis and hence we conclude that the mean number of errors made by these three classes of students are not equal.

3.4 TWO-WAY ANOVA

In two-way ANOVA a study variable is compared over three or more groups, controlling for another variable. The grouping is taken as one factor and the control is taken as another factor. The grouping factor is usually known as Treatment. The control factor is usually called Block. The accuracy of the test in two-way ANOVA is considerably higher than that of the one-way ANOVA, as the additional factor, block is used to reduce the error variance.

In two-way ANOVA, the data can be represented in the following tabular form.

Blocks

Groups or Treatments	1	2	3	…	m	$x_{i.}$
1	x_{11}	x_{12}	$x_{13.}$	⋯	x_{1m}	$x_{1.}$
2	x_{21}	x_{22}	$x_{2.}$	⋯	x_{2m}	$x_{2.}$
3	x_{31}	x_{32}	$x_{3.}$	⋯	x_{3m}	$x_{3.}$
⋮	⋮	⋮	⋮	⋮	⋮	⋮
k	x_{k1}	x_{k2}	x_{k3}	⋯	x_{km}	$x_{k.}$
$x_{.j}$	$x_{.1}$	$x_{.2}$	$x_{.3}$	⋯	$x_{.m}$	G

We use the following notations.

x_{ij} - i^{th} treatment value from the jth block, $i = 1,2, ..., k; j = 1,2, ..., m$.

The ith treatment total - $x_{i.} = \sum_{j=1}^{m} x_{ij}$, $i = 1, 2, ..., k$

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